Muhammad Subhan Shahid

Aspiring AI / Machine Learning Engineer
Lahore, Pakistan

+92 317 4103599 | subhanshahid1501@gmail.com | github.com/subhans1501 | https://subhan-portfolio-8zms.vercel.app/

AI-focused Artificial Intelligence undergraduate with hands-on experience in machine learning, NLP, and model deployment. Skilled in transformer architectures, reinforcement learning, and automating ML pipelines. Proven ability to design end-to-end AI systems and lead complex university projects. Actively seeking AI/ML internships.

EDUCATION

National University of Computer and Emerging Sciences Aug 2023 – Present
Bachelors in Artificial Intelligence
Unique College, Lahore *2021 - 2023*
Intermediate in Computer Science (ICS)
Unique School, Lahore *2019 - 2021*
Matriculation in Computer Science

TECHNICAL SKILLS

- **AI / ML:** NLP, Reinforcement Learning, Neural Architecture Search, MLOps, LLMs, Prompt Engineering
- **Frameworks:** MLflow, Gradio, Streamlit, PyTorch, TensorFlow, Hugging Face, Scikit-learn
- **Programming:** Python, C, C++, SQL
- **Tools:** Git, GitHub, Linux, Makefiles, VS Code

PROJECTS

Agentic Dialogue Response Generation

- Transitioned standard Seq2Seq text generation into an Agentic AI framework to bridge conversational AI and autonomous tool execution.
- Fine-tuned a quantized Llama-3 (8B) model using LoRA/PEFT to enhance contextual dialogue and integrated it within a LangChain/LangGraph architecture to autonomously query external tools and vector databases.
- Utilized a dual-dataset approach, leveraging DailyDialog for multi-turn conversational fluency and ToolBench to train the model in agentic function-calling and JSON API structures.

Conversational AI with RAG and Lightweight Agentic Tool Integration

- Developed a hybrid NLP conversational AI system that combines text generation, semantic retrieval, and agentic decision-making to seamlessly handle multi-turn dialogues.
- Engineered a Retrieval-Augmented Generation (RAG) pipeline integrating a FAISS vector database to dynamically retrieve context from a custom knowledge base, significantly reducing hallucination and improving factual accuracy over baseline models.

- Built an intent-driven framework capable of evaluating user queries to selectively trigger and utilize external APIs when task-oriented answers are required.

Fake Face (Deepfake) Detection System

- Developed a computer vision model utilizing Convolutional Neural Networks (CNNs) to accurately distinguish between real and AI-generated human faces.
- Preprocessed image datasets and optimized model weights to achieve high classification accuracy in identifying deepfake artifacts.

Fake News Detection System

- Built a classification pipeline using transformer models (BERT, RoBERTa, DistilBERT) to process and detect fabricated news.
- Fine-tuned models on custom datasets to generate high-accuracy classifications.

Text Summarization using Encoder-Decoder Models (BART, T5)

- Implemented abstractive text summarization using Transformer-based encoder-decoder architectures.
- Processed the arXiv dataset on Kaggle to generate concise and coherent summaries.

Phishing Email Detection & Deployment

- Trained and evaluated Logistic Regression, FNN, RNN, and LSTM models for accurately identifying phishing emails.
- Deployed the highest-performing NLP model through an interactive Gradio/Streamlit user interface.

MLOps Pipeline Automation

- Automated machine learning workflows utilizing Makefiles and Git version control for efficient model deployment.
- Tracked experiment metrics and evaluated model performance using MLflow for an Iris classification model.

SKILLS

- **AI & Machine Learning:** NLP, Computer Vision, Reinforcement Learning, Neural Architecture Search, MLOps, LLMs.
- **Frameworks & Tools:** MLflow, Gradio, Streamlit, PyTorch, TensorFlow, Hugging Face, Git, GitHub.
- **Programming Languages:** Python, C, C++, SQL.

INTERESTS

- Agentic AI, Machine Learning Research, Autonomous Systems, Reinforcement Learning.